

A Technical Introduction to Gaussian process

$$f(x) \sim \mathcal{GP}(m(x), k(x, x'))$$

Section 1

or $E[\exp(i(X_t - \mu))] = \exp(-\sigma^2/2)$. The numbers σ_j and μ_j can be shown to be the covariances and means of the variables in the process.

Section 2

Stationary case For a stationary Gaussian process $X = (X_t)_{t \in \mathbb{R}}$, some conditions on its spectrum are sufficient for sample continuity, but fail to be necessary. A necessary and sufficient condition, sometimes called Dudley–Fernique theorem, involves the function σ defined by

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As such, almost all sample paths of a mean-zero Gaussian process with positive definite kernel K will lie outside of the Hilbert space $H(K)$.

Section 4

RKHS structure and Gaussian process Let f be a mean-zero Gaussian process $\{X_t; t \in T\}$ with a non-negative definite covariance function K and let R be a symmetric and positive semidefinite function. Then, there exists a Gaussian process X which has the covariance R . Moreover, the reproducing kernel Hilbert space (RKHS) associated to R coincides with the Cameron–Martin theorem associated space $R(H)$ of X , and all the spaces $R(H)$, H_X , and $H(K)$ are isometric. From now on, let $H(R)$ be a reproducing kernel Hilbert space with positive definite kernel R . Driscoll's zero-one law is a result characterizing the sample functions generated by a Gaussian process:

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Then the constraint F_X can be fulfilled by choosing $f(x) = G_X(g(x))$, where $g(x) \sim \mathcal{GP}(\mu_g, K_g)$ is modelled as a Gaussian process, and finding G_X such that

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$f(x) = G_X g \sim \mathcal{GP}(G_X \mu_g, G_X K_g G_X^T)$.

Section 7

Some history. Sufficiency was announced by Xavier Fernique in 1964, but the first proof was published by Richard M. Dudley in 1967. Necessity was proved by Michael B. Marcus and Lawrence Shepp in 1970. There exist sample continuous processes X such that $I(\sigma) = \infty$; they violate condition (*). An example found by Marcus and Shepp is a random lacunary Fourier series

Section 8

$\text{var}[X(t)] = E[|X(t) - E[X(t)]|^2] < \infty$ for all $t \in T$.

Section 9

Brownian motion as the integral of Gaussian processes A Wiener process (also known as Brownian motion) is the integral of a white noise generalized Gaussian process. It is not stationary, but it has stationary increments. The Ornstein–Uhlenbeck process is a stationary Gaussian process. The Brownian bridge is (like the Ornstein–Uhlenbeck process) an example of a Gaussian process whose increments are not independent. The fractional Brownian motion is a Gaussian process whose covariance function is a generalisation of that of the Wiener process.

Section 10

Bayesian neural networks are a particular type of Bayesian network that results from treating deep learning and artificial neural network models probabilistically, and assigning a prior distribution to their parameters. Computation in artificial neural networks is usually organized into sequential layers of artificial neurons. The number of neurons in a layer is called the layer width. As layer width grows large, many Bayesian neural networks reduce to a Gaussian process with a closed form compositional kernel. This Gaussian process is called the Neural Network Gaussian Process (NNGP) (not to be confused with the Nearest Neighbor Gaussian Process). It

allows predictions from Bayesian neural networks to be more efficiently evaluated, and provides an analytic tool to understand deep learning models.

Section 11

Matérn: $K_{\text{Matern}}(x, x') = \frac{2^{1-\nu} \Gamma(\nu)}{\Gamma(\nu)} (2\sqrt{d} \|x - x'\|)^\nu K_\nu(2\sqrt{d} \|x - x'\|)$ $\{ \displaystyle K_{\text{Matern}}(x, x') = \frac{2^{1-\nu} \Gamma(\nu)}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu} d}{\|x - x'\|} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu} d}{\|x - x'\|} \right) \}$

Section 12

Stationarity For general stochastic processes strict-sense stationarity implies wide-sense stationarity but not every wide-sense stationary stochastic process is strict-sense stationary. However, for a Gaussian stochastic process the two concepts are equivalent. A Gaussian stochastic process is strict-sense stationary if and only if it is wide-sense stationary.

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where $\xi_1, \eta_1, \xi_2, \eta_2, \dots$ $\{ \displaystyle \xi_1, \eta_1, \xi_2, \eta_2, \dots \}$ are independent random variables with standard normal distribution; frequencies $0 < \lambda_1 < \lambda_2 < \dots$ $\{ \displaystyle 0 < \lambda_1 < \lambda_2 < \dots \}$ are a fast growing sequence; and coefficients $c_n > 0$ $\{ \displaystyle c_n > 0 \}$ satisfy $\sum_n c_n < \infty$. $\{ \text{style} \sum_n c_n < \infty \}$ The latter relation implies

Section 14

Linearly constrained Gaussian processes For many applications of interest some pre-existing knowledge about the system at hand is already given. Consider e.g. the case where the output of the Gaussian process corresponds to a magnetic field; here, the real magnetic field is bound by Maxwell's equations and a way to incorporate this constraint into the Gaussian process formalism would be desirable as this would likely improve the accuracy of the algorithm. A method on how to incorporate linear constraints into Gaussian processes already exists: Consider the (vector valued) output function $f(x)$ $\{ \displaystyle f(x) \}$ which is known to obey the linear constraint (i.e. $F^T X$ $\{ \displaystyle \mathcal{F}^T X \}$ is a linear operator)